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IMBALANCED HISTOPATHOLOGY IMAGE CLASSIFICATION USING DEEP FEATURE GRAPH ATTENTION NETWORK

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ABSTRACT

Recent deep learning studies have shown great progress in metastasis detection over histopathology whole slide images (WSIs). As WSIs are extremely large, most of existing studies adopt patch level analysis, leveraging spatial context to enhance patch-wise classification. However, class imbalance in patch distribution may result in an exceedingly large number of false-negatives, thereby worsening the WSI level classification performance. In this paper, we propose a novel framework for classification in class imbalanced datasets, which adopts a graph attention network to capture feature dependent interactions and a minority preferred inference mechanism for patch-level classification. Our experiments on CAMELYON16 show that the proposed method substantially improves detection of the minority class (tumour) under a highly imbalanced class distribution.

***Index Terms*—** Metastasis Detection, Graph Attention Networks, Intra-Class Feature Enhancement.

1. INTRODUCTION

Accurate detection of cancer and metastasis in whole-slide images (WSIs) is important for escalating treatments and increasing patient survival rate. However, it is time-consuming for pathologists to thoroughly examine WSIs; and the results normally suffer from high inter- and intra-pathologist variability. Recently, deep learning-based approaches have shown great progress in various WSI analysis tasks, such as classification, detection and segmentation. Due to the extremely large size of WSIs, it is infeasible to feed an entire WSI into the model. Most previous studies [1, 2] have adopted a patch-based approach: 1) analyse patches cropped from tissue regions; 2) aggregate patch-level results to obtain slide level results. Therefore, the performance of patch-level analysis has a significant impact on the WSI level analysis.

To improve patch-level analysis, current methods often incorporate spatial correlation between patches, rather than processing each patch individually. For instance, the neural conditional random field (NCRF) [3] method attaches a conditional random field (CRF) layer on top of the features extracted by convolutional neural networks (CNNs). Recently, NCRF is further improved by selecting more representative neighbouring patches [4]. However, the above methods still face the problem that positive (tumour) patches can be overwhelmed by negative (normal) patches, which leads to a high false-negative rate. This is largely due to the class imbalance issues in WSI datasets, i.e., the positive areas are always much smaller than the negative areas (Fig. 1). A few attempts have been made to solve the class imbalance issue in a histopathology dataset, wherein the re-sampling methods are normally used [5, 6]; however, the re-sampled data may disturb the intrinsic data distribution. Other methods employ model ensemble [7] or two-stage training [8] to reduce the number of false positives, which can either create extra computational overheads or complicate the training procedure.

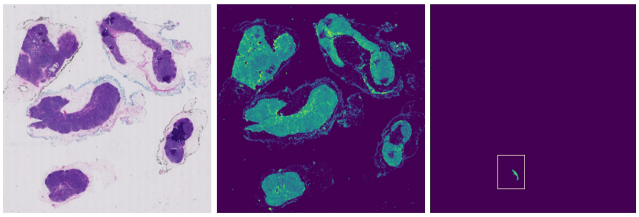

Fig. 1: Class imbalance can be prevalent in a WSI (a). Compared to the majority class (normal) in (b), the minority class (tumour) in (c) only takes up a small proportion.

To address the class imbalance issue in patch classification, we modify the current CNN-based classifiers. First, as the pooling layers in CNNs cannot capture the relationship between features with different receptive fields, we resort to graph attention networks (GATs) [9]. GATs capture the feature inter-dependencies between different receptive fields, which enhance intra-class interactions and weaken inter-class interactions. Second, to further reduce the impact of the difference in the number of samples between classes, inspired by the standard multiple instance learning assumption [10], we introduce a minority preferred inference procedure. Compared to existing re-sampling and ensemble methods,

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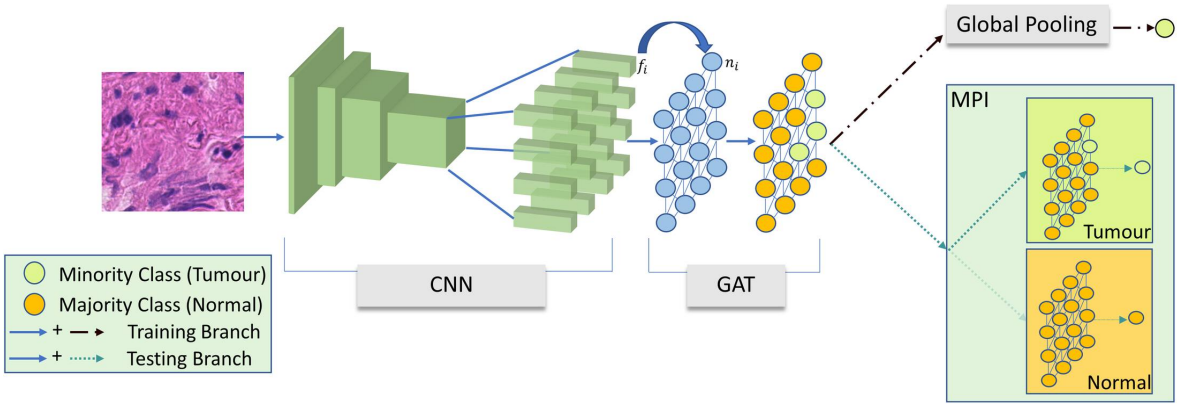

Fig. 2: The overall structure of the proposed DeepGAT model. The CNN-extracted features are fed into a graph attention network (GAT) layer to obtain node-level predictions. During training, a global pooling layer is used to generate patch-level predictions, whereas in the inference stage, the designed minority preferred inference (MPI) mechanism is used to generate patch-level predictions.

our method achieves superior performance with negligible extra computational cost. Experimental results on the public dataset CAMELYON16 [11] show that our method effectively reduces false-negatives and achieves higher F1 scores.

2. METHODS

The overall structure of our proposed deep feature graph attention network (DeepGAT) is shown in Fig. 2. We first obtain deep features by classic CNNs. Then, we construct a graph, with each node representing an element on the extracted feature and each edge representing the inter-dependency between adjacent elements. The graph is fed into a GAT module for feature aggregation, wherein the output of each node represents the probability of its corresponding receptive field being normal or tumour. In the inference phase, we adopt the MPI mechanism, i.e., a patch is classified as positive (tumour) if at least one node is classified as positive.

2.1. Deep Feature Graph Attention Network

Our proposed DeepGAT is composed of a CNN and a GAT, where the CNN is used to extract deep features and the GAT captures feature inter-dependence relationships. To be more specific, an input to the CNN is a patch cropped from WSI, and the activation map $f \in \mathcal{R}^{h\times w\times c} = (f_1, f_2, \cdots, f_{w\times h})$ before the average pooling layer of the CNN is then used as deep features fed into the GAT module. Here, different feature vectors $f_i (1 \leq i \leq w \times h)$ represent different effective receptive fields. To feed the features f into the GAT module, we construct a graph with nodes $n = (n_1, n_2, \cdots, n_{w\times h})$, where n_i has feature f_i, and there is an edge e_{ij} connecting n_i and n_j if their features f_i and f_j are spatial neighbours in the feature map. The graph is processed using two consecutive GAT layers where node features are updated by:

$$\mathbf{f}_i' = \alpha_{i,i}\mathbf{\Theta}\mathbf{f}_i + \sum_{j\in\mathcal{N}(i)} \alpha_{i,j}\mathbf{\Theta}\mathbf{f}_j \tag{1}$$

$$\alpha_{i,j} = \frac{\exp\left(\text{LeakyReLU}\left(\mathbf{a}^\top[\mathbf{\Theta}\mathbf{f}_i \,\|\, \mathbf{\Theta}\mathbf{f}_j]\right)\right)}{\sum_{k\in\mathcal{N}(i)\cup\{i\}} \exp\left(\text{LeakyReLU}\left(\mathbf{a}^\top[\mathbf{\Theta}\mathbf{f}_i \,\|\, \mathbf{\Theta}\mathbf{f}_k]\right)\right)} \tag{2}$$

Here $\alpha_{i,j}$ is the attention weight indicating the importance of feature f_j to feature f_i; Θ is a linear transformation; and $\mathbf{a}$ is a single-layer feedforward neural network. We denote the updated node features from the last GAT layer as $f^{GAT} = (f_1^{GAT}, f_2^{GAT}, \cdots, f_{w\times h}^{GAT})$, where f_i^{GAT} is the updated feature of node i. Then, f^{GAT} is fed into a global mean pooling layer for the final prediction, with a cross-entropy loss. The GAT in our model is to capture feature dependent relationships between $f_1, f_i, \cdots, f_{w\times h}$, therefore enhances intra-class interactions and weakens inter-class interactions. Our experiments show the effectiveness of GAT under our imbalanced setting and details are presented in Section 3.

2.2. Minority Preferred Inference Mechanism

To further reduce the impact of disparity between sample sizes of the majority class and the minority class, we propose to use the MPI mechanism. This is inspired by the standard multiple instance learning assumption [12], which assumes each object is a bag of instances; and states that all negative bags contain only negative instances, while positive bags contain at least one positive instance. Following this idea, instead of feeding f^{GAT} into the global mean pooling layer, we use the following formula to obtain the patch prediction ($Y_{f^{GAT}}$) during the inference stage:

$$Y_{f^{GAT}} = \begin{cases} 1 & if \quad \exists f_i^{GAT} \quad Y_{f_i^{GAT}} = 1 \\ 0 & if \quad \forall f_i^{GAT} \quad Y_{f_i^{GAT}} = 0 \end{cases} \tag{3}$$

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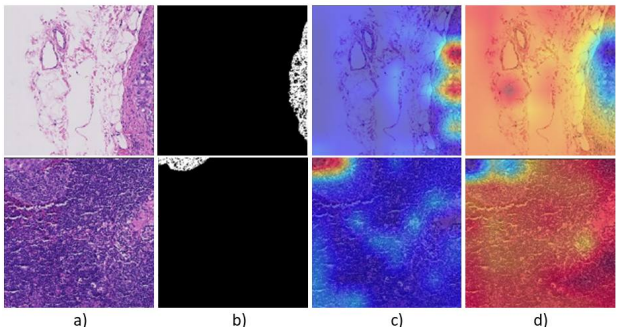

Fig. 3: Visualisation of Grad-CAM generated class activation maps using different models. (a) Image patch; (b) Annotated tumour regions; (c) and (d) the class activation maps generated by $\text{DeepGAT}_{\text{ResNet50}}$ and ResNet50, respectively. In (c) and (d), red colours indicate that the models have stronger confidence in predicting the positive class (tumour) regions.

We first obtain multiple predictions ($Y_{f_1^{GAT}}$, $Y_{f_2^{GAT}}$,$\cdots$, $Y_{f_{w\times h}^{GAT}}$) where $Y_{f_i^{GAT}}$ is the prediction on node feature f_i^{GAT} and the patch is classified as positive if at least one $Y_{f_i^{GAT}}$ is positive. This inference mechanism benefits positive classification because any one positive $Y_{f_i^{GAT}}$ can lead to a patch level positive which reduces the number of false-negatives. However, since the MPI mechanism is hard to be parameterised as a differentiable function, we still adopt global mean pooling for training.

3. EXPERIMENTS AND RESULTS

3.1. Dataset and Implementation

In this study, we conduct patch-level classification on the publicly available WSI dataset from CAMELYON16 [11] for automated detection of breast cancer metastasis in lymph node sections. Since the pixel annotations are provided in CAMELYON16, this allows us to perform patch-level classification. We first apply the Otsu algorithm to filter out the background regions of each slide and then densely extract patches of size 256×256 pixels at $5\times$ magnification level. Patches containing cancer metastasis are labelled as positive (tumour) and others are labelled as negative (normal). In the end, there are 64,828 patches, where 89.3% of them (57,942) are positive and 10.7% of them (6866) are negative. We then randomly split them at the slide level into training, validation and test sets with a ratio of 80:10:10.

We use ResNet50 and DenseNet121 as feature extractors as they are superior on natural image classification tasks. We replace the last pooling and fully connected layers with GAT. The model is trained for 60 epochs using a NVIDIA RTX 3090 with a batch size of 32 and a learning rate of 0.0002. We develop the using PyTorch.

	PR	SP	R	F1	$\text{NP}_{(\text{Million})}$
ResNet	0.888	0.987	0.819	0.852	25.5
DenseNet	0.889	0.988	0.837	0.862	8.0
ResNet+CRF	0.801	0.984	0.785	0.792	25.5
DenseNet+CRF	0.806	0.986	0.816	0.811	8.0
$\text{DeepGAT}_{\text{ResNet}}$	**0.951**	**0.994**	**0.930**	**0.940**	25.6
$\text{DeepGAT}_{\text{DenseNet}}$	**0.943**	**0.994**	**0.927**	**0.935**	7.5

Table 1: The classification performance and number of parameters (NP) of different CNN backbones and graphical models.

	PR	SP	R	F1
ResNet	0.888	0.987	0.819	0.852
ResNet + CB [13]	0.890	0.988	0.825	0.856
ResNet + LDAM [14]	0.905	0.989	0.850	0.876
ResNet + AUG	0.900	0.993	0.876	0.888
ResNet + DS	0.895	0.957	0.930	0.913
ResNet + SMOTE	0.921	0.962	0.891	0.920
ResNet + GAT	0.941	0.993	0.918	0.930
$\text{DeepGAT}_{\text{ResNet50}}$	**0.951**	**0.994**	**0.930**	**0.940**

Table 2: Comparison between different class re-balancing and inference mechanisms.

3.2. Experimental Results

Since we are dealing with a highly imbalanced dataset, classification metrics, such as accuracy, may not reflect the performance of the model. Instead, we measure the precision (PR), recall (R), specificity (SP) and F1-score (F1) which are more commonly used for imbalanced classifications. Table 1 shows the classification results. Particularly, CNN classifiers, such as ResNet and DenseNet, are not able to accurately predicate the minority class as their precision and recall are much lower than their specificity. Moreover, using CRF results in worse performance, as spatial correlation between neighbouring patches might weaken the feature representations of minority (positive) cases. Compared to baseline CNN classifiers, DeepGAT, with an negligible number of extra parameters or even less parameters, consistently produces higher precision and recall, indicating that our model accurately detects the minority class (positive).

We also investigated the use of different approaches to alleviate class imbalance, including the loss re-weighting methods (with class-balanced loss (CB) [13] and label distribution-aware margin loss (LDAM) [14], respectively), the strong augmentation on minority classes (AUG) method, and the data re-sampling methods (down-sampling (DS) and up-sampling based on Synthetic Minority Oversampling Technique (SMOTE) [15]). Both CB and LDAM show marginal improvement. We argue that the power of the baseline classifier is limited to effectively discriminate inter-class features. Extra augmentations on the minority class, including horizon-

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tal/vertical flipping, rotation and random sharpness/contrast adjustment, leads to over 3% F1-score improvement. Some further improvements can be achieved using data re-sampling methods but they either suffer from over-fitting on the minority class or require extra data pre-processing to generate synthetic samples. Our DeepGAT achieves significant F1-score improvement ($\sim 9\%$) without extra data pre-processing.

We further compare our proposed MPI technique with the average pooling-based inference mechanism (ResNet+GAT) in which the patch prediction is made by averaging the prediction over feature vectors (same as the training phase). From Table 2, we can observe that ResNet+GAT and DeepGAT are the best two methods, which shows the effectiveness of using GAT under our highly imbalanced setting. In addition, DeepGAT outperforms ResNet+GAT, which means the MPI meachmisn can not only decrease false-negatives leading to a higher recall but also reduce false-positives resulting in a higher precision

For qualitative results, we show the Grad-CAM [16] visualisation of the ResNet50 baseline and DeepGAT in Fig. 3. From the results, it is clear that adding the proposed GAT layer on top of the CNN feature extractor produces much better performance in detecting tumour regions. This demonstrates that DeepGAT can effectively enhance the intra-class feature and benefits the performance of tasks under imbalanced class settings.

4. CONCLUSIONS

In this paper, we proposed a novel framework (DeepGAT) to enhance feature learning on WSI patches under a highly imbalanced setting. We incorporate the graph attention network to better capture the feature dependent relationships between different receptive fields. We also design a minority preferred inference mechanism which further addresses the significant imbalance between the majority and minority classes. Experimental studies on CAMELYON16 show that DeepGAT can achieve substantial performance improvement ($\sim 9\%$) in detecting the minority (tumour) class with negligible extra or less computational cost. This motivates us to investigate novel graph neural networks to further enhance feature learning and make our method more applicable on general image tasks for future studies.

5. COMPLIANCE WITH ETHICAL STANDARDS

A public dataset was used and no ethical approval was required.

6. ACKNOWLEDGEMENTS

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